§4.3.

P1. Solution: First, let us find all equilibrium solutions.

$$f\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x-x^3-xy^2\\2y-y^5-yx^4\end{pmatrix}$$

$$\Rightarrow\begin{cases}x-x^3-xy^2=0 \Rightarrow x(1-x^2\ y^2)=0 & -① \\ 2y-y^5-yx^4=0\Rightarrow y(2-y^4-x^4)=0 & -②\end{cases}$$

$① \Rightarrow x=0$ or $x^2+y^2=1$

$② \Rightarrow y=0$ or $x^4+y^4=2$

i) $x=0$ & $y=0$

ii) $x=0$ & $x^4+y^4=2 \Rightarrow x=0,\ y=\pm\sqrt[4]{2}$

iii) $x^2+y^2=1$ & $y=0 \Rightarrow x=\pm1,\ y=0$

iv) $x^2+y^2=1$ & $x^4+y^4=2 \Rightarrow$ no solution.

Thus, this equation has 5 equilibrium solutions: $\underline{x(t)=0,\ y(t)=0}$, $\underline{x(t)=0,\ y(t)=\sqrt[4]{2}}$, $\underline{x(t)=0,\ y(t)=-\sqrt[4]{2}}$, $\underline{x(t)=1,\ y(t)=0}$, $\underline{x(t)=-1,\ y(t)=5}$.

Using Method I to determine their stabilities

I) $x(t)=0,\ y(t)=0$: set $u=x,\ v=y$.

$$\frac{d}{dt}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}u-u^3-uv^2\\2v-v^5-vu^4\end{pmatrix}=\begin{pmatrix}1&0\\0&2\end{pmatrix}\begin{pmatrix}u\\v\end{pmatrix}+\begin{pmatrix}-u^3-uv^2\\-v^5-vu^4\end{pmatrix}$$

$A=\begin{pmatrix}1&0\\0&2\end{pmatrix}\Rightarrow\lambda_1=1,\ \lambda_2=2$

$\Rightarrow$ unstable.

II) $x(t)=0,\ y(t)=\sqrt[4]{2}$: set $u=x,\ v=y-\sqrt[4]{2}$

$$\frac{d}{dt}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}u-u^3-u(v+\sqrt[4]{2})^2\\2(v+\sqrt[4]{2})-(v+\sqrt[4]{2})^5-(v+\sqrt[4]{2})u^4\end{pmatrix}$$

$$=\begin{pmatrix}1-\sqrt{2}&0\\0&-8\end{pmatrix}\begin{pmatrix}u\\v\end{pmatrix}+(\text{higher-order terms})$$

$A=\begin{pmatrix}1-\sqrt{2}&0\\0&-8\end{pmatrix}\Rightarrow\lambda_1=1-\sqrt{2}<0$, $\lambda_2=-8<0$

$\Rightarrow$ asympototically stable.

III: $x(t)=0,\ y(t)=-\sqrt[4]{2}$: set $u=x,\ v=y+\sqrt[4]{2}$

$$\frac{d}{dt}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}u-u^3-u(v-\sqrt[4]{2})^2\\2(v-\sqrt{2})-(v-\sqrt[4]{2})^5-(v+\sqrt[4]{2})u^4\end{pmatrix}$$

$$=\underbrace{\begin{pmatrix}1-\sqrt{2}&0\\0&-8\end{pmatrix}}_{A}\begin{pmatrix}u\\v\end{pmatrix}+(\text{higher-order terms})$$

$\Rightarrow\lambda_1=1-\sqrt{2}<0$, $\lambda_2=-8<0$

$\Rightarrow$ asymptotically stable.

IV: $x(t)=1,\ y(t)=0$: $u=x-1,\ v=y$

$$\frac{d}{dt}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}(u+1)-(u+1)^3-(u+1)v^2\\2v-v^5-v(u+1)^4\end{pmatrix}$$

$$=\begin{pmatrix}-2&0\\0&1\end{pmatrix}\begin{pmatrix}u\\v\end{pmatrix}+(\text{higher-order terms})$$

$\Rightarrow A=\begin{pmatrix}-2&0\\0&1\end{pmatrix}\Rightarrow\lambda_1=-2,\ \lambda_2=1$

$\Rightarrow$ unstable.

V: $x(t)=-1,\ y(t)=0$: $u=x+1,\ v=y$

$$\frac{d}{dt}\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}(u-1)-(u-1)^3-(u-1)v^2\\2v-v^5-v(u-1)^4\end{pmatrix}$$

$$=\underbrace{\begin{pmatrix}-2&0\\0&\end{pmatrix}}_{A}\begin{pmatrix}u\\v\end{pmatrix}+(\text{higher-order terms})$$

$\Rightarrow\lambda_1=-2,\ \lambda_2=1$

$\Rightarrow$ unstable.

P3. Solution: First, let us find all equilibrium solutions

$$f\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x^2+y^2-1\\2xy\end{pmatrix}$$

$$\Rightarrow\begin{cases}x^2+y^2-1=0 \Rightarrow x^2+y^2=1\\ 2xy=0\Rightarrow x=0 \text{ or } y=0\end{cases}$$

i) $x=0$ & $x^2+y^2=1 \Rightarrow x=0,\ y=\pm 1$

ii) $y=0$ & $x^2+y^2=1 \Rightarrow x=\pm 1,\ y=0$

Thus, this equation has 4 equilibrium solutions: $\underline{x(t)=0,\ y(t)=1}$, $\underline{x(t)=0,\ y(t)=-1}$, $\underline{x(t)=1,\ y(t)=0}$, $\underline{x(t)=-1,\ y(t)=0}$.

Using method II to determine their stability.

$$A=\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} \quad \text{where } f_1=x^2+y^2-1,\ f_2=2xy$$

$$=\begin{bmatrix} 2x & 2y \\ 2y & 2x \end{bmatrix}$$

I) $x(t)=0,\ y(t)=1$:

$$A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \lambda_1=1,\ \lambda_2=-1$$

$\Rightarrow$ unstable

II) $x(t)=0,\ y(t)=-1$:

$$A=\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \Rightarrow \lambda_1=1,\ \lambda_2=-1$$

$\Rightarrow$ unstable.

III) $x(t)=1,\ y(t)=0$:

$$A=\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \lambda_1=\lambda_2=2$$

$\Rightarrow$ unstable.

(IV) $x(t)=-1,\ y(t)=0$:

$$A=\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \Rightarrow \lambda_1=\lambda_2=-2$$

$\Rightarrow$ asymptotically stable.

P5. solution: Let us first find all equilibrium solutions

$$f\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} \tan(x+y) \\ x+x^3 \end{pmatrix}$$

$$\left\{\begin{array}{l} \tan(x+y)=0 \\ x+x^3=0 \Rightarrow x(1+x^2)=0 \Rightarrow x=0 \end{array}\right\} \Rightarrow$$

$\tan y=0 \Rightarrow y=k\pi$ where k is an integer. Thus, this equation has infinitely many equilibrium solutions

$\underline{x(t)=0,\ y(t)=k\pi}$, $k=0,\pm 1,\pm 2,\cdots$

Let us use method II to determine their stability.

$f_1=\tan(x+y),\quad f_2=x+x^3$

$$A=\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}=\begin{bmatrix} 1+\tan^2(x+y) & 1+\tan^2(x+y) \\ 1+2x^2 & 0 \end{bmatrix}$$

$x(t)=0,\ y(t)=k\pi$:

$$A=\begin{bmatrix} 1+\tan^2(k\pi) & 1+\tan^2(k\pi) \\ 1 & 0 \end{bmatrix}$$

$$=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow P(\lambda)=\lambda^2-\lambda-1 \Rightarrow \lambda_1=\frac{1+\sqrt{5}}{2},\ \lambda_2=\frac{1-\sqrt{5}}{2}$$

$\Rightarrow$ unstable for all integer k.

P.9. Solution:

$$f\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} f_1 \\ f_2 \end{pmatrix}=\begin{pmatrix} e^{x+y}-1 \\ \sin(x+y) \end{pmatrix}$$

clearly, $f\begin{pmatrix} 0 \\ 0 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ so that $\underline{x(t)=0,\ y(t)=0}$ is an equilibrium solution.

$$A=\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}=\begin{bmatrix} e^{x+y} & e^{x+y} \\ \cos(x+y) & \cos(x+y) \end{bmatrix}$$

$x(t)=0,\ y(t)=0$:

$$A=\begin{bmatrix} e^0 & e^0 \\ \cos 0 & \cos 0 \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow P(\lambda)=\lambda^2-2\lambda$$

$\Rightarrow \lambda_1=0,\ \lambda_2=2$

$\Rightarrow$ unstable.

P11. Solution: $f\binom{x}{y} = \begin{pmatrix} \cos y - \sin x - 1 \\ x - y - y^2 \end{pmatrix} \begin{matrix} \to f_1 \\ \to f_2 \end{matrix}$

clearly $f\binom{0}{0} = \binom{0}{0}$, so that $\underline{x(t)=0,\ y(t)=0}$

is an equilibrium solution.

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} -\cos x & -\sin y \\ 1 & -1-2y \end{bmatrix}$$

$x(t)=0,\ y(t)=0$:

$$A = \begin{bmatrix} -\cos 0 & -\sin 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$$

$\Rightarrow P(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda+1)^2$

$\Rightarrow \lambda_1 = \lambda_2 = -1$

$\Rightarrow$ asymptotically stable